And it is the "tail" that determines the convergence (divergence) of a sequence (V. Imp)

- Thus, we have arrived at the following

Conclusion: Elimination, addition, and any other change of a finite number of terms of a sequence do not affect either its convergence or its limit (if the sequence is convergent). → V. Imp. But this needs proper proof

Reader: I guess that an elimination of an infinite number of terms (for ex, every other term) must not affect the convergence of a sequence either.

Author: Here you must be very careful. If an initial sequence is convergent, an elimination of an infinite number of its terms (provided that the number of the remaining terms is also infinite) does not affect either convergence or the limit of the sequence. (Needs proof)

- If, however, an initial sequence is divergent, an elimination of an infinite number of its terms may, in certain cases, convert the sequence into a convergent one.

- Seq 10: $1, \frac{2}{3}, \frac{1}{3}, \frac{4}{5}, \frac{1}{5}, \frac{6}{7}, \frac{1}{7}, \frac{8}{9}, \ldots\ldots$ → Divergent

- Eliminate from this sequence all the terms with even serial numbers, you will get the convergent sequence.

$$1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \frac{1}{9}, \frac{1}{11}, \frac{1}{13}, \ldots\ldots$$

- Suppose we form from a given convergent sequence two new convergent sequences. The first new sequence will consist of the terms of the initial sequence with odd serial numbers, while the second will consist of the terms with even serial numbers. What do you think are the limits of these new sequences?

Reader: It is easy to prove that the new sequences will have the same limit as the initial sequence. (Needs proof)

Author: You are right.